

solution 7

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BS24B006

THE WATERMARK BEHIND EACH IMAGE IS THE PROOF THAT I AM NOT COPYING

(I got zero in lots of assignments even though i did myself 😭)

The output for each file can be checked again by running these commands, i have not given output and code inside this document for many assignments previously because only code was asked in assignments (codes are again in the github link

https://github.com/specapoorv/BT3040_BioInfo_Lab_assignments)

```
git clone
https://github.com/specapoorv/BT3040_BioInfo_Lab_assignm
ents.git

cd assignment_7
python3 q{x}.py #replace x with question number for eg 1
```

q.1

assignment_7 >  q1.py

```
1 s1 = "RATPTRWPVGC FNRPWTKWSYDEALDGIKAAGYAWTGLLTASKPSLHHATATPEYLAALKQKSRH
2 s2 = "AAAVMMGLAAIGAAIGIGILGGKFLEGAARQPDLIPLLRTQFFIVMGLVDAIPMIAVGLGLYVMF
3 s3 = "AADVSAAVGATGQSGMTYRLGLSWDWDKSWWQTSTGRLTGYWDAGYTYWEGGDEGAGKHSLSFAP
4
5 aa = "ACDEFGHIKLMNPQRSTVWY"
6
7 for x in [("seq1", s1), ("seq2", s2), ("seq3", s3)]:
8     print(x[0])
9     l = len(x[1])
10    for a in aa:
11        c = x[1].count(a)
12        p = (c / l) * 100
13    print(a, round(p, 2))
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

...

 Python

+ v





...





spec-base% /usr/bin/python3 /home/specapoorv/bioinfo/assignment_7/q1.py

seq1

A 17.91

C 1.49

D 2.99

E 2.99

F 1.49

G 5.97

H 4.48

I 1.49

K 7.46

L 8.96

M 0.0

N 1.49

P 7.46

Q 1.49

R 5.97

S 5.97

T 10.45

V 1.49

W 5.97

Y 4.48

seq2

A 19.12

C 0.0

D 2.94

E 1.47

F 5.88

G 14.71

 0  0  Live Share

Spaces: 4 UTF-8 LF {} Python Python 3.10.12



```

T 4.48
seq2
A 19.12
C 0.0
D 2.94
E 1.47
F 5.88
G 14.71
H 0.0
I 11.76
K 1.47
L 13.24
M 7.35
N 0.0
P 4.41
Q 2.94
R 2.94
S 0.0
T 1.47
V 8.82
W 0.0
Y 1.47
seq3
A 10.6
C 0.0
D 5.96
E 3.97
F 5.3
G 15.23

```

0 0 Live Share

```

Y 1.47
seq3
A 10.6
C 0.0
D 5.96
E 3.97
F 5.3
G 15.23
H 1.32
I 5.3
K 3.97
L 5.96
M 0.66
N 3.31
P 2.65
Q 3.31
R 3.31
S 9.93
T 4.64
V 5.3
W 3.97
Y 5.3
spec-base% 

```

0 0 Live Share Spaces: 4

amino acid	seq1 (%)	seq2 (%)	seq3 (%)
A	17.91	19.12	10.60

amino acid	seq1 (%)	seq2 (%)	seq3 (%)
C	1.49	0.00	0.00
D	2.99	2.94	5.96
E	2.99	1.47	3.97
F	1.49	5.88	5.30
G	5.97	14.71	15.23
H	4.48	0.00	1.32
I	1.49	11.76	5.30
K	7.46	1.47	3.97
L	8.96	13.24	5.96
M	0.00	7.35	0.66
N	1.49	0.00	3.31
P	7.46	4.41	2.65
Q	1.49	2.94	3.31
R	5.97	2.94	3.31
S	5.97	0.00	9.93
T	10.45	1.47	4.64
V	1.49	8.82	5.30
W	5.97	0.00	3.97
Y	4.48	1.47	5.30

seq1 has mostly ala and thr also high pro and leu, mix of polar and nonpolar residue but met is zero and cys only one maybe mixed secondary structure domain

seq2 is having crazy amount of hydrophobic amino acid like a, g, l, i, v and met also there, charged stuff is very less and no c h n s w at all so it looks like transmembrane helix or membrane protein

seq3 got very high gly and ser with balanced polar and nonpolar aa everywhere, lots of aromatic residue also there and zero cys looks like

normal globular soluble protein or beta sheet loop region

```
assignment_7 > q2.py > ...
1  w = {
2      "A": 85,
3      "C": 115,
4      "D": 130,
5      "E": 145,
6      "F": 160,
7      "G": 70,
8      "H": 150,
9      "I": 125,
10     "K": 145,
11     "L": 125,
12     "M": 143,
13     "N": 130,
14     "P": 110,
15     "Q": 140,
16     "R": 170,
17     "S": 100,
18     "T": 115,
19     "V": 110,
20     "W": 200,
21     "Y": 175,
22 }
23
24 s1 = "RATPTRWPVGCENRPWTKWSYDEALDGIKAAGYAWTGLLTASKPSLHHATATPEYLAALKQKSRHAA"
25 s2 = "AAAVMMGLAAIGAAIGILGGKFLEGAARQPDLIPLLRTOFFIVMGLVDAIPMIAVGLGLYVMFAVA"
26 s3 = "AADVSAAVGATGQSGMTYRLGLSWDWDKSWWQTSTGRLTGYWDAGYTYWEGGDEGAGKHSLSFAPVFVYEFAGDSIKPFIEAGIGVAAFSGTR"
27
28 for i, s in enumerate([s1, s2, s3], 1):
29     tot = 0
30     for x in s:
31         tot += w[x]
32     print("seq", i, tot)
```

Y 5.3

```
spec-base%
spec-base% /usr/bin/python3 /home/specapoorv/bioinfo/assignment_7/q2.py
seq 1 8315
seq 2 7735
seq 3 18153
spec-base%
```

q3

```
    "Y": 5.3,
}

for i, c in enumerate([c1, c2, c3], 1):
    d_a = sum((c[k] - ga[k]) ** 2 for k in ga) ** 0.5
    d_b = sum((c[k] - gb[k]) ** 2 for k in gb) ** 0.5
    grp = "A" if d_a < d_b else "B"
    print(f"seq{i}: dist_a={d_a:.2f}, dist_b={d_b:.2f} -> Group {grp}")
```

(only giving core logic of code, please if you dont see full code check the github link, here for your convenience

https://github.com/specapoorv/BT3040_BioInfo_Lab_assignments/blob/main/assignment_7/q3.py

)

```
spec-base% /usr/bin/python3 /home/specapoorv/bioinfo/assignment_7/q3.py
seq1: dist_a=15.70, dist_b=15.73 -> Group A
seq2: dist_a=19.94, dist_b=20.68 -> Group A
seq3: dist_a=10.94, dist_b=9.59 -> Group B
spec-base%
```

q4

```
aa = "ACDEFGHIKLMNPQRSTVWY"

s1 = "RATPTRWPVGCNRPWTKWSYDEALDGIKAAGYAWTGLLTASKPSLHHATATPEYLAALKQKSRHAA"
s2 = "AAAVMMGLAAIGAAIGIGILGGKFLEGAARQPDLIPLLRTQFFIVMGLVDAIPMIAVGLGLYVMFAVA"
s3 = "AADVSAAVGATGQSGMTYRLGLSWDWDKSWWTSTGRLTGYWDAGYTYWEGGDEGAGKHSLSFAPVFVYEFAGDSIKPFIEAGIGVAAFSGTR"

for sn, s in [("seq1", s1), ("seq2", s2), ("seq3", s3)]:
    print("=" * 40)
    print(sn)
    n = len(s)
    f = {x: s.count(x) for x in aa}
    nij = {}
    for a1 in aa:
        for a2 in aa:
            nij[(a1, a2)] = 0
    for i in range(n - 1):
        p = (s[i], s[i + 1])
        nij[p] += 1

    pa = {}
    pb = {}
    pc = {}

    for a1 in aa:
        for a2 in aa:
            cnt = nij[(a1, a2)]
            # (a)  $N_{ij} * 100 / (N_i + N_j)$ 
            if f[a1] + f[a2] > 0:
                pa[(a1, a2)] = (cnt * 100.0) / (f[a1] + f[a2])
            else:
                pa[(a1, a2)] = 0.0

            # (b)  $N_{ij} * 100 / (N - 1)$ 
            pb[(a1, a2)] = (cnt * 100.0) / (n - 1)

            # (c)  $N_{ij} * 100 / (N_i * N_j)$ 
            if f[a1] * f[a2] > 0:
                pc[(a1, a2)] = (cnt * 100.0) / (f[a1] * f[a2])
            else:
                pc[(a1, a2)] = 0.0

    print("--- 20x20 table pref (b) ---")
    print(" " + " ".join([f"{x:>5}" for x in aa]))
```

1. Sequence 1

20x20 Residue Pair Preference Matrix: $[N_{ij} \cdot 100 / (N - 1)]$
(%)

A	C	D	E	F	G	H	I	K	L
A	4.55	0.00	0.00	0.00	0.00	1.52	0.00	0.00	0.
C	0.00	0.00	0.00	0.00	1.52	0.00	0.00	0.00	0.
D	0.00	0.00	0.00	1.52	0.00	1.52	0.00	0.00	0.
E	1.52	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.
F	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.
G	0.00	1.52	0.00	0.00	0.00	0.00	0.00	1.52	0.
H	3.03	0.00	0.00	0.00	0.00	0.00	1.52	0.00	0.
I	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.
K	1.52	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.
L	1.52	0.00	1.52	0.00	0.00	0.00	1.52	0.00	1.
M	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.
N	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.
P	0.00	0.00	0.00	1.52	0.00	0.00	0.00	0.00	0.
Q	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.
R	1.52	0.00	0.00	0.00	0.00	0.00	1.52	0.00	0.
S	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.
T	3.03	0.00	0.00	0.00	0.00	1.52	0.00	0.00	1.
V	0.00	0.00	0.00	0.00	0.00	1.52	0.00	0.00	0.
W	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.
Y	1.52	0.00	1.52	0.00	0.00	0.00	0.00	0.00	0.

Top 10 Residue Pairs (Sequence 1)

Rank	Preference (a): [$N_{ij} \cdot 100 / (N_i + N_j)$]	Preference (b): [$N_{ij} \cdot 100 / (N - 1)$]	Preference (c): [$N_{ij} \cdot 100 / (N_i \cdot N_j)$]
1	CF (50.00%)	AA (4.55%)	CF (100.00%)
2	FN (50.00%)	AT (4.55%)	FN (100.00%)
3	DE (25.00%)	AL (3.03%)	DE (25.00%)
4	EY (20.00%)	HA (3.03%)	GC (25.00%)
5	GC (20.00%)	TA (3.03%)	GI (25.00%)
6	GI (20.00%)	TP (3.03%)	NR (25.00%)
7	NR (20.00%)	WT (3.03%)	VG (25.00%)
8	VG (20.00%)	AG (1.52%)	IK (20.00%)
9	YD (20.00%)	AS (1.52%)	KQ (20.00%)
10	WT (18.18%)	AW (1.52%)	PV (20.00%)

2. Sequence 2

20x20 Residue Pair Preference Matrix: [$N_{ij} \cdot 100 / (N - 1)$]
(%)

A	C	D	E	F	G	H	I	K	L
A	7.46	0.00	0.00	0.00	0.00	0.00	0.00	4.48	0.
C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.
D	1.49	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.
E	0.00	0.00	0.00	0.00	0.00	1.49	0.00	0.00	0.
F	1.49	0.00	0.00	0.00	1.49	0.00	0.00	1.49	0.
G	2.99	0.00	0.00	0.00	0.00	1.49	0.00	2.99	1.
H	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.
I	1.49	0.00	0.00	0.00	0.00	4.48	0.00	0.00	0.
K	0.00	0.00	0.00	0.00	1.49	0.00	0.00	0.00	0.
L	1.49	0.00	0.00	1.49	0.00	2.99	0.00	1.49	0.

A	C	D	E	F	G	H	I	K	L
M	0.00	0.00	0.00	0.00	1.49	2.99	0.00	1.49	0.
N	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.
P	0.00	0.00	1.49	0.00	0.00	0.00	0.00	0.00	0.
Q	0.00	0.00	0.00	0.00	1.49	0.00	0.00	0.00	0.
R	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.
S	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.
T	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.
V	1.49	0.00	1.49	0.00	0.00	1.49	0.00	0.00	0.
W	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.
Y	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.

Top 10 Residue Pairs (Sequence 2)

Rank	Preference (a): [Nij·100/(Ni+Nj)]	Preference (b): [Nij·100/(N-1)]	Preference (c): [Nij·100/(Ni·Nj)]
1	RT (33.33%)	AA (7.46%)	RT (50.00%)
2	TQ (33.33%)	GL (5.97%)	TQ (50.00%)
3	VM (27.27%)	AI (4.48%)	KF (25.00%)
4	RQ (25.00%)	AV (4.48%)	RQ (25.00%)
5	GL (21.05%)	IG (4.48%)	PD (16.67%)
6	KF (20.00%)	VM (4.48%)	QP (16.67%)
7	PD (20.00%)	GA (2.99%)	YV (16.67%)
8	QP (20.00%)	GI (2.99%)	QF (12.50%)
9	AA (19.23%)	IP (2.99%)	LE (11.11%)
10	IP (18.18%)	LG (2.99%)	LY (11.11%)

3. Sequence 3

20x20 Residue Pair Preference Matrix: $[N_{ij} \cdot 100 / (N - 1)]$ (%)

A	C	D	E	F	G	H	I	K	L
A	2.00	0.00	0.67	0.00	0.67	4.00	0.00	0.67	0.
C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.
D	0.67	0.00	0.00	0.67	0.00	0.67	0.00	0.00	0.
E	0.67	0.00	0.67	0.00	0.67	1.33	0.00	0.00	0.
F	2.00	0.00	0.00	0.67	0.00	0.00	0.00	0.67	0.
G	2.00	0.00	2.00	0.00	0.00	0.67	0.00	1.33	0.
H	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.
I	0.00	0.00	0.00	1.33	0.00	1.33	0.67	0.00	0.
K	0.00	0.00	0.00	0.00	0.67	0.00	0.67	0.67	0.
L	0.00	0.00	0.00	0.00	0.67	1.33	0.00	0.00	1.
M	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.
N	0.67	0.00	0.67	0.00	0.67	0.67	0.00	0.00	0.
P	0.00	0.00	0.00	0.00	0.67	0.00	0.00	0.67	0.
Q	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.
R	0.67	0.00	0.00	0.00	0.00	0.00	0.00	0.67	0.
S	0.67	0.00	0.00	0.00	0.67	1.33	0.00	0.67	0.
T	0.00	0.00	0.00	0.00	0.00	2.00	0.00	0.00	0.
V	0.67	0.00	0.00	0.00	0.67	2.00	0.00	0.00	0.
W	0.00	0.00	2.00	0.67	0.00	0.00	0.00	0.00	0.
Y	0.00	0.00	0.00	0.67	0.00	0.00	0.00	0.00	0.

Top 10 Residue Pairs (Sequence 3)

Rank	Preference (a): [Nij·100/(Ni+Nj)]	Preference (b): [Nij·100/(N-1)]	Preference (c): [Nij·100/(Ni·Nj)]
1	WD (20.00%)	AG (4.00%)	MT (14.29%)
2	AG (15.38%)	AA (2.00%)	KH (8.33%)
3	IE (14.29%)	FA (2.00%)	HY (6.25%)
4	RL (14.29%)	GA (2.00%)	IH (6.25%)
5	YW (14.29%)	GD (2.00%)	WD (5.56%)
6	LK (13.33%)	GL (2.00%)	PN (5.00%)
7	TY (13.33%)	SL (2.00%)	QP (5.00%)
8	FA (12.50%)	TG (2.00%)	RL (4.44%)
9	KH (12.50%)	VG (2.00%)	GM (4.35%)
10	MT (12.50%)	WD (2.00%)	IE (4.17%)

q5

```
# table values:
# row 30: Hgm
# row 19: Ca
# row 14: Et

p = {
    "A": {"hgm": 13.85, "ca": 20.0, "et": 1.90},
    "D": {"hgm": 11.61, "ca": 26.0, "et": 1.52},
    "C": {"hgm": 15.37, "ca": 25.0, "et": 2.04},
    "E": {"hgm": 11.38, "ca": 33.0, "et": 1.54},
    "F": {"hgm": 13.93, "ca": 46.0, "et": 1.86},
    "G": {"hgm": 13.34, "ca": 13.0, "et": 1.90},
    "H": {"hgm": 13.82, "ca": 37.0, "et": 1.76},
    "I": {"hgm": 15.28, "ca": 39.0, "et": 1.95},
    "K": {"hgm": 11.58, "ca": 46.0, "et": 1.37},
    "L": {"hgm": 14.13, "ca": 35.0, "et": 1.97},
    "M": {"hgm": 13.86, "ca": 43.0, "et": 1.96},
    "N": {"hgm": 13.02, "ca": 28.0, "et": 1.56},
    "P": {"hgm": 12.35, "ca": 22.0, "et": 1.70},
    "Q": {"hgm": 12.61, "ca": 36.0, "et": 1.52},
    "R": {"hgm": 13.10, "ca": 55.0, "et": 1.48},
    "S": {"hgm": 13.39, "ca": 20.0, "et": 1.75},
    "T": {"hgm": 12.70, "ca": 28.0, "et": 1.77},
    "V": {"hgm": 14.56, "ca": 33.0, "et": 1.98},
    "W": {"hgm": 15.48, "ca": 61.0, "et": 1.87},
    "Y": {"hgm": 13.88, "ca": 46.0, "et": 1.69},
}

s1 = "RATPTRWPGCFNRPWTKWSYDEALDGIIKAAGYAWTGLLTASKPSLHHATATPEYLAALKQKSRHAA"
s2 = "AAAVMMGLAAIGAAIGIGILGGKFLEGAARQPDLIPLLRTOFFIVMGLVDAIPMIAVGLGLYVMFAVA"
s3 = "AADVSAAVGATGQSGMTYRLGLSWDWDKSWQTSTGRITGYWDAGYTYWEGGDEGAGKHSLSFAPVFVYEFAGDSIKPFIEAGIGVAAFSGTR"

for i, s in enumerate([s1, s2, s3], 1):
    n = len(s)
    tot_h = sum(p[x]["hgm"] for x in s)
    tot_ca = sum(p[x]["ca"] for x in s)
    tot_et = sum(p[x]["et"] for x in s)
    print(
        f"seq{i}: hgm={tot_h/n:.3f}, ca={tot_ca/n:.2f}, et={tot_et/n:.2f} (len={n})"
    )
```

```
> /usr/bin/python3 /home/specapoorv/bioinfo/assignment_7/q5.py
seq1: hgm=13.353, ca=32.18, et=1.76 (len=67)
seq2: hgm=13.772, ca=30.40, et=1.86 (len=68)
seq3: hgm=13.419, ca=30.57, et=1.77 (len=151)
~/bioinfo main ?2 >
```

seq1 has lower hydrophobicity 13.353 but highest helical contact area 32.18 because of bulky charged and aromatic residues like arg lys trp, looks like typical water exposed surface globular chain

seq2 has the highest hydrophobicity 13.772 and highest energy 1.86 due to huge amounts of hydrophobic ala ile leu val packing together, strongly indicates a transmembrane segment

seq3 has middle values with hgm 13.419 and ca 30.57, well balanced polar and non-polar mix with high glycine keeping it flexible, shows a standard soluble globular domain